

The Omega Axiom

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The Omega Axiom: Closed-Phase Unified Framework with Z-Phase Dynamics
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Abstract

We present the **Omega Axiom**, a non-Lagrangian, closed-phase framework unifying energy, inertia, gravitation, and electromagnetic phenomena. By introducing **Z-phase**, **A-wave**, and **B-effect**, the framework links local phase compression events to global cosmological dynamics, producing a dynamic Λ without invoking dark energy or spacetime curvature. We provide a fully self-consistent model integrating local phase oscillations and global FRW expansion.

Keywords

Omega Axiom; Z-phase; A-wave; B-effect; Closed Phase; Dynamic Lambda; FRW Cosmology

1 Introduction

Conventional physics treats inertia and gravitation as fundamental forces or geometric necessities. The Omega Axiom proposes a radical simplification: *energy arises from phase compression in closed topologies, and apparent forces are residuals of phase desynchronization*. Electromagnetism is treated as the sole fundamental field, whose closed-phase dynamics yield macroscopic phenomena, including inertia and gravitation.

2 The Omega Axiom

Axiom (): Energy arises from logarithmic phase compression within a closed topological boundary. No fundamental force exists; all apparent forces are residuals of boundary phase misalignment.

$$E = m_p \cdot I \cdot r \tag{1}$$

where m_p is phase mass, $I = 1.50$ is the universal compression constant, and r is the closure radius.

3 Vortex Structure

3.1 Single (Open) Vortex

Unstable phase leakage; compression fluctuates around $I = 1.50$.

3.2 Paired (Closed) Vortex

Forms a closed synchronization loop; index stabilizes at

$$I_{\text{pair}} = 3.0$$

yielding residual inertial and gravitational effects. Inertia is interpreted as resistance to desynchronization.

4 A-Wave: Phase-Layer Oscillation

$$A(\phi) = \ln \frac{r_{n+1}}{r_n}, \quad r_{n+1}/r_n \approx \frac{\delta Z_{\text{local}}}{\langle Z \rangle_{\Omega}} \quad (2)$$

Predicts atmospheric standing-wave nodes (e.g., Venus, Jupiter) without thermal explanations.

5 B-Effect: Phase Shear Emission

$$B(x, t) = \frac{B_0}{1 + e^{-\alpha(\Phi_{\text{closed}}(x, t) - \Phi_{\text{crit}})}} \quad (3)$$

Describes luminous or energetic excess at boundaries of closed-phase systems; smooth sigmoid replaces Heaviside for observational fit.

6 Z-Phase and Dynamic Lambda

6.1 Local Definition

$$Z(x, t) = \left. \frac{\partial \Phi}{\partial t} \right|_{\text{closed}}, \quad \delta \rho_Z(x, t) = f(Z(x, t) - Z_{\text{crit}}) \quad (4)$$

f is a smoothed sigmoid function; nonzero Z corresponds to active phase transmission (A-wave).

6.2 Global FRW Integration

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = \frac{8\pi G}{3} \left[\rho_{\text{matter}}(t) + \frac{1}{V} \int_V \delta\rho_Z(x, t) d^3x \right] \quad (5)$$

$$\Lambda_{\text{dyn}}(t) = 8\pi G \langle \rho_{\text{tot}}(x, t) \rangle_\Omega \quad (6)$$

6.3 Time Evolution Feedback

$$\dot{\Lambda}_{\text{dyn}}(t) = 8\pi G \frac{1}{V} \int_V \frac{\partial}{\partial t} f(Z(x, t) - Z_{\text{crit}}) d^3x \quad (7)$$

7 Gravitation as Residual Phase Alignment

$$G_{\text{eff}}(r) \propto \frac{\ln I \cdot \langle Z(x, t) \rangle_r}{r^2} \quad (8)$$

Apparent inverse-square attraction emerges from averaged local Z-phase gradients, no spacetime curvature required.

8 Observational Consistency

- Galaxy rotation curves $\rightarrow$ large-scale Z-phase coherence
- Gravitational lensing $\rightarrow$ phase-path refraction
- Helium abundance $\rightarrow$ irreversible phase condensation
- FRW expansion $\rightarrow$ dynamic $\Lambda_{\text{dyn}}(t)$ from Z-phase activity

9 Discussion

The Omega Axiom replaces additive force frameworks with multiplicative phase logic. Non-Lagrangian formulation avoids variational paradoxes; all dynamics are constrained by closed-phase conditions and averaging.

10 Conclusion

Inertia, gravitation, and electromagnetic phenomena admit a unified description via closed-phase compression with universal logarithmic index $I = 1.50$. Z-phase, A-wave, and B-effect connect local activity to global dynamics, generating a dynamic Λ without invoking dark energy.

A Stability of Compression Index

Linear perturbation analysis with Z-phase feedback confirms $I = 1.50$ as a stable fixed point under boundary-preserving transformations.

References

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